

Ride Sharing Optimization

ECS 311 Project Presentation

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1. Introduction
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Problem Statement

- **Assigning Passenger Groups to Vehicles:** Groups must be allocated to vehicles with a fixed capacity.
- **Fair Cost Allocation:** The fare of each route segment is split equally among all passengers riding that segment.
- **Minimizing the Maximum Individual Fare:** The goal is to ensure that the highest fare any individual pays, over all shared segments, is as low as possible.

Literature Review

- *A Ride Sharing System for University Community (Olaniyan et al.)*
Objective: Proposes a university-specific ride-sharing platform to reduce commute issues and promote sustainability.
Inspiration: Emphasizes tailored solutions with features like geo-tracking and user verification, ideal for closed-campus systems.
- *Optimization for Dynamic Ride-Sharing: A Review (Agatz et al.)*
Objective: Surveys optimization techniques for real-time ride-sharing under operational constraints.
Inspiration: Provides a toolkit of optimization strategies (e.g., heuristics, simulations) and highlights the importance of scalability.
- *Cost-Sharing Mechanism Design for Ride-Sharing*
Objective: Designs fair, incentive-compatible cost-sharing schemes for dynamic ride-sharing.
Inspiration: Enables stable and fair pricing models to prevent misuse and enhance user trust.

Route-Based Affinity Scoring : Groups passengers by similarity in journey routes.

Timing-Based Affinity Scoring: Refines groups based on departure time preferences.

Multi-Vehicle Distribution (Future work): Optimizes group savings when group size exceeds vehicle capacity.

Route-Based Affinity Scoring

City Map as a Tree-Graph

- Nodes represent locations.
- Edges represent roads between locations.
- **IISER Bhopal** is the root node (common source/destination).
- Tree structure ensures a **unique path** between any two nodes.

Passenger Route Representation

- Each passenger P_n has a route:

$$r_n = \langle a_1^{(n)}, a_2^{(n)}, \dots, a_{k_n}^{(n)} \rangle$$

- Route is a sequence of nodes from/to the institute.

Route-Based Affinity Scoring Contd...

Route Affinity Score (ρ_{ij})

- Measures overlap in route between passengers P_i and P_j .
- Defined as:

$$\rho_{ij} = \begin{cases} \frac{f_{\text{fare}}(\text{LCP}(r_i, r_j))}{f_{\text{fare}}(r_i)}, & \text{if } \text{LCP}(r_i, r_j) \in \{r_i, r_j\} \\ 0, & \text{otherwise} \end{cases}$$

- $\text{LCP}(r_i, r_j)$: Longest Common Prefix of r_i and r_j .
- $f_{\text{fare}}(r)$ denotes the fare charged to travel along route r
- Asymmetry reflects **fare-sharing potential**—greater overlap implies more savings.

Modeling Passenger Time Preferences

Goal: Model each passenger's preferred departure time as a *Normal Distribution* such that a proportion m_{range} (e.g., 80%) of the probability mass lies between t_{min} and t_{max} .

Step 1: Time Convenience Distribution

Given:

- t_{min} : earliest preferred departure time
- t_{max} : latest preferred departure time
- m_{range} : desired area under the curve between t_{min} and t_{max} (default: 0.8)

Compute:

$$\mu = \frac{t_{\text{min}} + t_{\text{max}}}{2}$$
$$z = \Phi^{-1} \left(\frac{1 + m_{\text{range}}}{2} \right), \quad \sigma = \frac{t_{\text{max}} - t_{\text{min}}}{2z}$$

Where Φ^{-1} is the inverse CDF (percent-point function) of the standard normal distribution.

Computing Time Affinity Score

Goal: Measure how much of *Passenger i*'s time preference distribution overlaps with *Passenger j*'s time window.

Step 2: Time Affinity Score

Let:

- (μ_i, σ_i) : mean and std of Passenger *i*'s time preference distribution
- $[t_{\min}^j, t_{\max}^j]$: Passenger *j*'s preferred time window

Compute the proportion of Passenger *i*'s probability mass that lies within Passenger *j*'s window:

$$\tau_{ij} = \min \left(1, \frac{\Phi_i \left(\frac{t_{\max}^j - \mu_i}{\sigma_i} \right) - \Phi_i \left(\frac{t_{\min}^j - \mu_i}{\sigma_i} \right)}{m_{\text{range}}} \right)$$

Where Φ is the CDF of the normal distribution.

Implementation Methodology

Objective: Optimize auto-rickshaw sharing to reduce fares and maximize ride-sharing efficiency.

Modeling the Map:

- Locations across Bhopal are represented as **nodes**.
- Roads connecting locations have **weights** equal to fare values.
- A **rooted tree** structure is constructed with IISER Bhopal as the root node.

User Input:

- Each passenger specifies their **source**, **destination**, and **preferred time**.
- One of the locations must be **IISER Bhopal**.

Route Search:

- For each user, the optimal route is retrieved using a **search algorithm** on the location tree.

Implementing Affinity Calculation and Fare Optimization

Affinity Computation:

- A **Route Affinity Matrix** and a **Time Affinity Matrix** are constructed.
- These are multiplied element-wise to form the **Final Affinity Matrix**.
- **Min-Max Scaling** is applied to normalize the values.

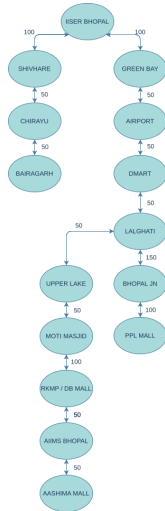
Clustering and Grouping:

- **Affinity Propagation Clustering** is applied to group passengers with similar routes and time preferences.
- Each group represents an **optimized shared ride**.

Fare and Savings Calculation:

- For each passenger, the **optimized shared fare** is computed.
- **Individual savings** are calculated compared to their original solo fare.

Bhopal Map and System Flowchart



Bhopal Map Representation

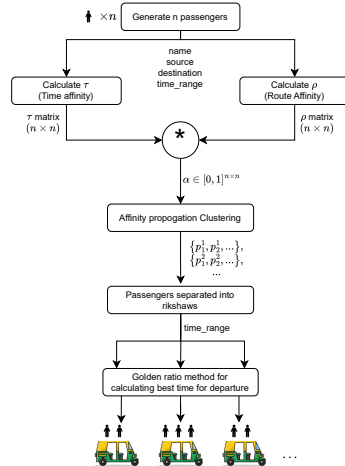


Figure: Flow chart

Ride Optimization Program Overview

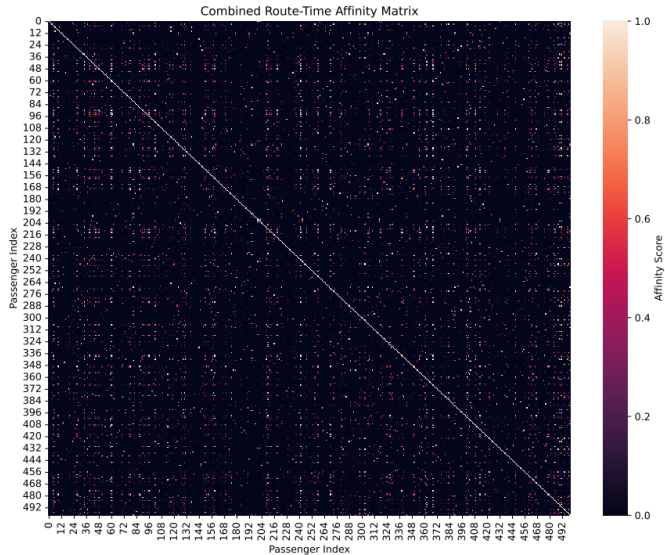
- **Objective:** Optimize shared transportation by pooling passengers with similar routes to reduce costs.
- **Inputs:** Passenger routes, original fares, and destinations.
- **Outputs:**
 - Optimized fare for each passenger
 - Total savings (in and %)
 - Recommended departure time
 - Summary stats: total fare, total passengers, total savings
- **Key Features:**
 - Matches passengers with overlapping routes
 - Minimizes fare per person and maximizes total savings
 - Tabular summaries for transparency
 - Ideal for shuttles, carpools, and group rides

Ride Optimization Outputs

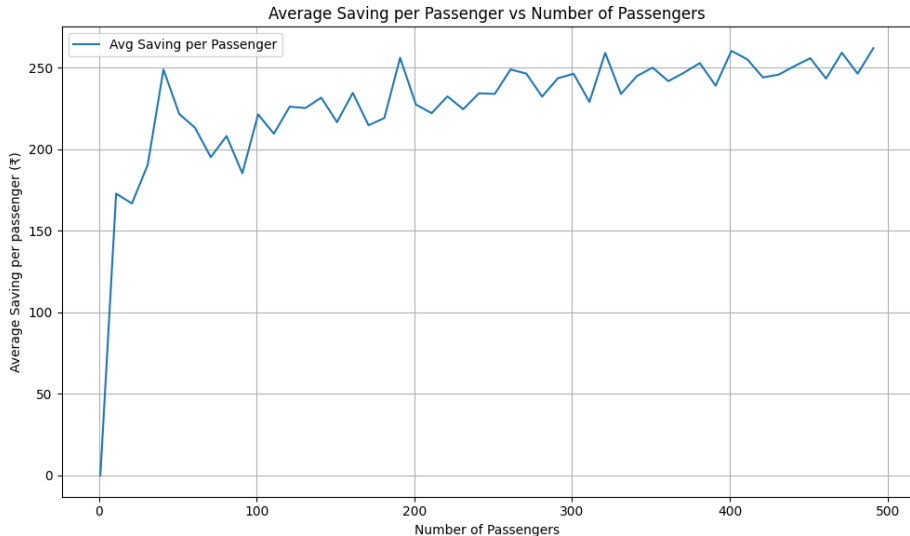
Detail	Value
Optimized Departure Time	09:09 24 Apr 2025
Total Passengers	7
Total Fare	₹200.00
Total Savings	₹1000.00

Passenger	Route	Original Fare	New Fare	Savings	Savings (%)
Passenger #126	IISER Bhopal → Bairagarh	₹200.00	₹33.33	₹166.67	83.33
Passenger #143	IISER Bhopal → Shivhare	₹100.00	₹16.67	₹83.33	83.33
Passenger #258	IISER Bhopal → Bairagarh	₹200.00	₹33.33	₹166.67	83.33
Passenger #410	IISER Bhopal → Bairagarh	₹200.00	₹33.33	₹166.67	83.33
Passenger #454	IISER Bhopal → Bairagarh	₹200.00	₹33.33	₹166.67	83.33
Passenger #467	IISER Bhopal → Bairagarh	₹200.00	₹33.33	₹166.67	83.33
Passenger #490	IISER Bhopal → Shivhare	₹100.00	₹16.67	₹83.33	83.33

Combined Affinity Matrix Plot



A comparative Study : Savings vs Number of Passengers



- **Ride-sharing** is a sustainable, cost-effective, and eco-friendly commuting practice.
- Particularly **beneficial for student communities**, reducing financial burden and promoting inclusivity.
- Encourages more travel, boosting **local economic growth**.
- Facilitates easy ride-sharing connections, **reducing plan cancellations** due to lack of coordination.
- Promotes environmental consciousness by **cutting down carbon emissions and pollution**.
- Empowers youth — the most active demographic — to engage in more social and recreational activities.

Future Work Scope

- **Optimal Group Splitting:** If a group exceeds the vehicle's capacity, it must be divided into smaller trips.
- **Privacy-preserving data collection** to learn user travel patterns (time, date, location).
- **Heuristic-based prediction** of travel patterns for better ride match suggestions.
- Exploration of **Deep Reinforcement Learning** using Markov Decision Processes to optimize multiple route options.
- **Mobile and Web App Development** targeted initially at IISER Bhopal, with plans for broader community rollout.
- Transition from manual planning (e.g., WhatsApp groups) to a **streamlined digital platform** for enhanced ease of use and reliability.
- Aiming for **greater networking, inclusivity, and prosperity** through smart, community-driven transportation solutions.